ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Congruence, Similarity & Geometrical Proof

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CONGRUENCE, SIMILARITY & GEOMETRICAL PROOF · CH4 · L8

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The Map

This lesson collapses Congruence, Similarity & Geometrical Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 11 website questions, 11 checked solutions, 4 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 12 Diagram NOT accurately drawn A $(2x + 9)$ cm $2x$ cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$CD = 1.5BE$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2HR Q12

Find the gradient
This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$CD = 1.5BE$$

Finish: (a) $ED = 5$ cm, (b) $x = 3.75$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 12 10 cm Diagram NOT accurately drawn E D C A B In the diagram, ABC and AED are straight lines. BE is parallel to CD AE = 10 cm and CD = 1.5 times BE (a) Work out the length of ED cm (2) AB = (2x + 5) cm and BC = (3x - 5) cm (b) Work out the value of x x = (2)

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 12 Diagram NOT accurately drawn A (2x + 9) cm 2x cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Evaluate fraction
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.